

Relational Algebra – Beginner’s Guide

with Syntax, Examples, and Explanations

1. Introduction

Relational Algebra (RA) is a **procedural query language** that operates on relations (tables) in a database. It provides a set of operations to retrieve and manipulate data logically.

2. Basic Operators

a) Selection (σ)

Used to **filter rows** based on a condition.

$$\sigma_{\text{condition}}(\text{Relation})$$

Example: Find employees who earn more than 50000.

$$\sigma_{\text{Salary} > 50000}(\text{Employee})$$

b) Projection (π)

Used to **select specific columns (attributes)**.

$$\pi_{\text{columns}}(\text{Relation})$$

Example: Display names and salaries of employees.

$$\pi_{\text{Name, Salary}}(\text{Employee})$$

c) Renaming (ρ)

Used to **rename a relation or attributes**.

$$\rho_{\text{NewName}}(\text{OldRelation}) \quad \text{or} \quad \rho_{\text{NewName}(A,B,C)}(\text{OldRelation})$$

3. Set Operations

a) Union ($\cup$)

Combines tuples from two relations (removes duplicates).

$$R \cup S$$

Example: Employees from two departments.

$$\text{Employee_IT} \cup \text{Employee_HR}$$

b) Intersection ($\cap$)

Returns tuples that exist in **both** relations.

$$R \cap S$$

Example: Employees who work in both Project A and Project B.

$$\text{Employee_A} \cap \text{Employee_B}$$

c) Difference / Minus ($-$)

Returns tuples in R but not in S.

$$R - S$$

Example: Employees in HR but not in IT.

$$\text{Employee_HR} - \text{Employee_IT}$$

4. Joins

a) Cartesian Product ($\times$)

Combines all tuples from R and S.

$$R \times S$$

Example:

$$\text{Employee} \times \text{Department}$$

b) Theta Join ($\bowtie_{\text{condition}}$)

Combines tuples based on a condition.

$$R \bowtie_{R.\text{attr} = S.\text{attr}} S$$

Example: Join Employee and Department where their IDs match.

$$\text{Employee} \bowtie_{\text{Employee.deptID} = \text{Department.deptID}} \text{Department}$$

c) Natural Join ($\bowtie$)

Automatically joins on all **common attributes**.

$$R \bowtie S$$

Example:

$$\text{Employee} \bowtie \text{Department}$$

d) Outer Joins

Keep unmatched tuples using:

- **Left Outer Join:** $R \ltimes S$

- **Right Outer Join:** $R \bowtie S$
 - **Full Outer Join:** $R \bowtie S$
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5. Grouping and Aggregation

a) Aggregation Functions

Common functions: COUNT, SUM, AVG, MIN, MAX

Example: Find average salary of employees.

$$\text{AVG}(\text{Salary})(\text{Employee})$$

b) Group By and Aggregate ($\mathcal{G}$)

Used to group tuples by some attributes and apply aggregate functions.

$$\mathcal{G}_{\text{GroupAttr}}^{\text{AggregateFunc}}(\text{Relation})$$

Example: Find average salary per department.

$$\mathcal{G}_{\text{deptID}}^{\text{AVG}(\text{Salary})}(\text{Employee})$$

c) Having Clause (Filter After Grouping)

Used to filter groups based on aggregate values.

Example: Departments with average salary greater than 60000.

$$\sigma_{\text{AVG}(\text{Salary}) > 60000}(\mathcal{G}_{\text{deptID}}^{\text{AVG}(\text{Salary})}(\text{Employee}))$$

6. Division ($\div$)

Used for queries like “find entities related to all values of another set”.

Example: Find employees who know *all* programming languages.

$$\text{Employee_Language} \div \text{Language}$$

7. Complex Query Example

Query: Find names of employees who work in a department located in the same city as their home address.

$$\pi_{\text{Employee.name, Department.deptName}} \left(\sigma_{\text{Employee.city} = \text{Department.city}} \left(\text{Employee} \bowtie_{\text{Employee.deptID} = \text{Department.deptID}} \text{Department} \right) \right)$$

8. Summary of Operators

Operator	Meaning
σ	Selection (Filter rows)
π	Projection (Select columns)
ρ	Rename relation/attributes
$\cup$	Union
$\cap$	Intersection
$-$	Difference (Minus)
$\times$	Cartesian Product
$\bowtie$	Join
$\mathcal{G}$	Group By with Aggregate
$\div$	Division

9. Aliasing (Renaming Tables Temporarily)

In relational algebra, we can give a relation a temporary name (alias) using the **rename operator** ρ . This is useful in self-joins or when simplifying long expressions.

$$\rho_t(\text{Response_Team})$$

Here, `Response_Team` is renamed as alias `t`. We can then use this alias in subsequent expressions.

Example: Select the `team_name` of all teams that specialize in “AI Defense”.

$$\pi_{t.team_name}(\sigma_{t.specialization='AI\,Defense'}(\rho_t(\text{Response_Team})))$$

Meaning: Rename the table `Response_Team` as `t`, filter rows where specialization is 'AI Defense', and project only the `team_name` column.

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